THREAT Landscape Of ATTACKS On SMARTPHONE SENSORS

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Ever wondered what black magic powers do smartphone apps such as Sky Maps have, for them to know the precise orientation of your phone while you gaze at the heavens above? What about ride-hail apps such as Uber that can generate such stunningly granular reporting on the drivers' braking and acceleration patterns, or apps like Metal Detector that help you hunt ghosts?

In the next few minutes, you will be ushered into the world of the humble trio of inertial measurement unit (IMU) sensors, or motion sensors, that is, accelerometers, gyroscopes and magnetometers. These unsung little workhorses lurk unsuspectingly on our phones' motherboards, furiously transducing and spitting data emanating from every sinew of motion and vibration around us, while quietly fueling an emergent wave of artificial intelligence (AI) that will influence and shape our lives deeply in the coming decade.

Now, the question is: Does your smartphone have these as well? In case you own a reasonably new smartphone, like me, chances are that dialing a USSD secret code will give you backdoor access to a screen that looks like the one shown in Fig. 2.

Woah! You did not realise there was an entire menagerie of sensors prowling in your phone, did you? You are certainly not alone! According to a recent survey, a staggering 61 per cent of smartphone users had never even heard of accelerometers and an even higher ratio (73 per cent) had not heard of gyroscopes.

Let me give you an idea as to how incredibly sensitive and fast these sensors are. A typical IMU sensor has a linear acceleration sensitivity as low as 0.00061 g/LSB (gravitational acceleration per least signal bit) and an angular rate sensitivity of 0.004375 degrees per sec/LSB. This can be sampled at the rate of 6664 samples per second. However, in order to save battery, the mobile operating system limits the sampling rate to 416 samples per second, which is still impressively high. These sensors are so incredibly sensitive that these can be used as a reliable alternative to an electromyogram (EMG) for tremor frequency assessment when it comes to the care of patients with a diagnosis of Parkinson's disease, Essential tremor, Holmes' tremor and even Orthostatic tremor.

These examples are part of this very exciting revolution happening in the personal healthcare space, where the emergence of the motion sensors-laden smartphone as the centrepiece of democratisation of healthcare-hardware has yielded incredible results in domains such as gait- and posture-related disorders treatment, geriatric care and treatment of neurodegenerative disorders (Fig. 3)

Besides healthcare, these motion sensors are driving incredible advances in

Smartphones are equipped with many different sensors that allow us to interact with our real-world surroundings. However, our growing dependence on smartphones is generating considerable new threats, owing to the army of sensors that are increasingly vulnerable to attack

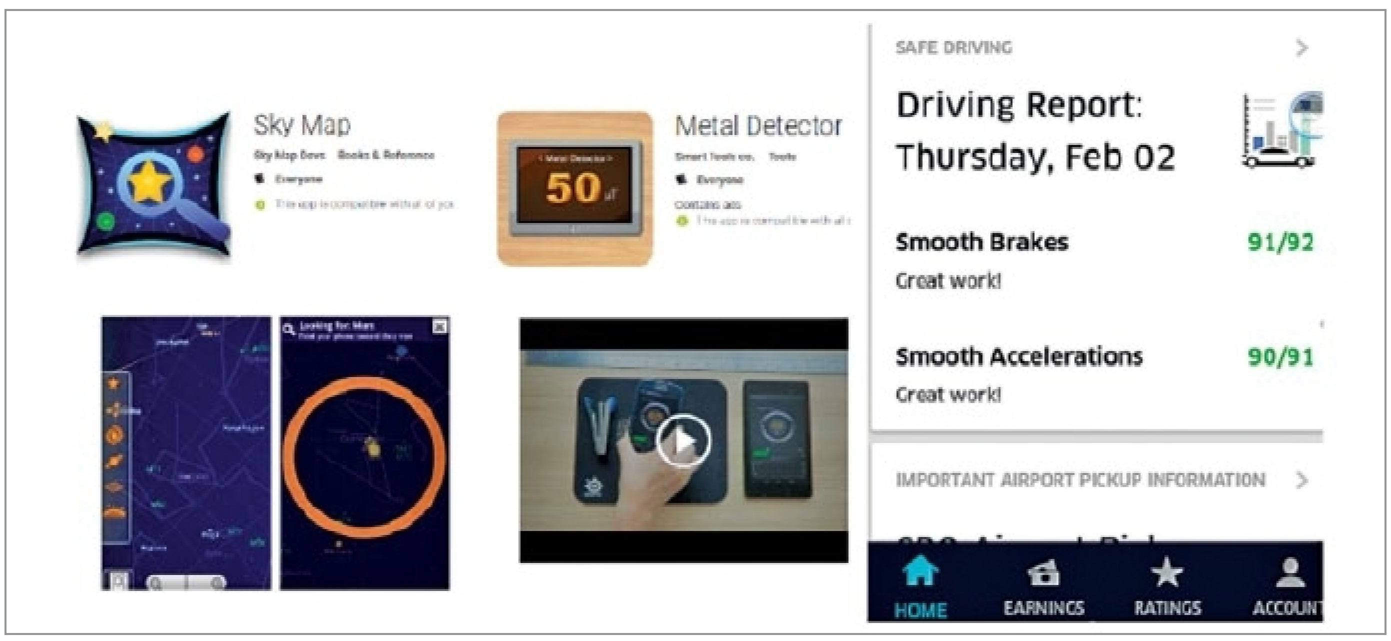

Fig. 1: Examples of popular apps that use inertial measurement unit sensors